INTRODUCTION

Prior lessons explored two sources of interest rate risk—reinvestment risk and price risk—and demonstrated how holding a bond for its Macaulay duration balances them. This lesson and those that follow extend that discussion by introducing measures of price risk. Two broad categories of such measures exist: those that assume underlying bond cash flows are certain and measure price sensitivity to changes in a bond's *own yield*, which are covered in these lessons, and those that introduce the possibility of a bond default and that measure price sensitivity to changes in a *benchmark* yield curve, which are covered in later lessons. This lesson will illustrate how the interest rate risk of a bond is a function of its features, including its time-to-maturity, coupon rate, and yield.

Most of the examples and exhibits used throughout the reading can be downloaded as a Microsoft Excel workbook. Each worksheet in the workbook is labeled with the corresponding example or exhibit number in the text.

Learning Module Overview

- Duration is a quantitative measure of interest rate risk. There are several duration measures, including those that measure a bond's price sensitivity to changes in its own yield-to-maturity and assume underlying cash flows are certain (yield duration) and those that measure changes in a benchmark yield curve, with less certain underlying cash flows (curve duration).
- Macaulay duration, modified duration, money duration, and the price value of a basis point (PVBP) are yield duration measures.
- Modified duration is the slope or first derivative of the price of a bond with respect to its yield-to-maturity, measuring the sensitivity of a bond's price to changes in its yield-to-maturity. Modified duration can be calculated using a bond's Macaulay duration and yield or through approximation.
- Money duration is an extension of modified duration and incorporates the size of the bond position in currency terms. Related to this measure is the price value of a basis point, which is an estimate of the change in the price of a bond for a 1 bp change in the bond's yield.
- Duration can be used to estimate the change in the price of a bond in response to a change in yield, but it assumes a linear relationship between price and yield even though, in fact, the relationship is nonlinear. This is most evident when estimating price changes for large changes in yield and for bonds with certain features.
- A bond's features, including its time-to-maturity, coupon rate, and yield-to-maturity, determine its duration. Duration, for a given bond, is not static and decreases as the bond approaches maturity.
- All else equal, a longer (shorter) time-to-maturity, a lower (higher) coupon rate, or a lower (higher) yield-to-maturity results in higher (lower) duration or higher (lower) interest rate risk.

Learning Module Self-Assessment

understanding of the learning module.

Match each of the following descriptions with the appropriate term:

Description	Term
Present-value-weighted average time-to-receipt of a bond's cash flows	i. Money duration
Used to estimate the percentage change in a bond's price for a given change in the bond's yield-to-maturity	ii. Price value per basis point
The product of the annualized modified duration and the full price of a bond	iii. Macaulay duration
Change in the full price of a bond for a 1 bp change in yield	iv. Modified duration

Hide Solution

Solution:

Present value weighted average time-to-receipt of a bond cash flows matches with iii. Macaulay duration.

Used to estimate the percentage change in a bond's price for a given change in the bond's yield-to-maturity matches with iv. Modified duration.

The product of the annualized modified duration and the full price of a bond matches with i. Money duration.

Change in the full price of a bond for a 1 bp change in yield matches with ii. Price value per basis point.

2. If all other characteristics of a bond are held constant, an increase in the bond's yield-to-maturity will *most likely* result in:

A. a decrease in the bond's modified duration.
B. no change in the bond's modified duration.
C. an increase in the bond's modified duration.

Hide Solution

Solution:

A is correct. An increase in the bond's yield-to-maturity will result in a lower modified duration because future cash flows are discounted at a higher rate, which reduces the present-value-weighted average of their time to receipt.

B is incorrect because duration is affected by the present value of the bond's cash flows, which, in turn, is affected by the bond's yield.

C is incorrect because an increase in the bond's yield-to-maturity reduces the present value of each cash flow and hence reduces the bond's duration.

3. Consider a bond with an annualized modified duration of 4, a coupon of 5%, and a price of 95. The expected change in the price of the bond, per 100 of par value, for a 50 bp increase in the bond's yield-to-maturity is *closest* to:

A. 4.75.
B. 3.80.
C. 1.90.

Hide Solution

Solution:

C is correct. The money duration is 380:

$$MoneyDur = AnnModDur \times PV^{Full}.$$

This results in a change in price of –1.90 for a 50 bp increase in the bond's yield:

$$\Delta PV^{Full} \approx -MoneyDur \times \Delta Yield.$$

$$\Delta PV^{Full} \approx -380 \times 0.005.$$

$$\Delta PV^{Full} \approx -1.90.$$

The change in price is negative for the increase in the yield-to-maturity.

4. Two bonds, Bond X and Bond Y, are identical except that Bond X has a lower coupon rate than Bond Y. Therefore:

A. the bonds will have identical modified duration.
B. Bond X will have a lower modified duration compared to Bond Y.
C. Bond Y will have a lower modified duration compared to Bond X.

Hide Solution

Solution:

C is correct. If two bonds are identical in all other respects (e.g., time-to-maturity, yield, etc.), the bond with the lower coupon rate will have the higher modified duration. This is because, compared to Bond Y, the present value of each of Bond X's coupon cash flows will make up less of the bond's price, while the contribution to the price of the bond from the face value will be greater.

5. Estimating the price value per basis point by dividing the difference in the price of a bond for a 0.0001 decrease and a 0.0001 increase by 2 ignores:

A. convexity of the bond's price–yield relationship.
B. concavity of the bond's price–yield relationship.
C. symmetry in the bond's price–yield relationship.

Hide Solution

Solution:

A is correct. This formulation of the price value of a basis point does not include an adjustment for convexity. This is important because the price change for increases and decreases in yields is asymmetric, owing to the convex relationship between price and yield.

6. The change in the price of a bond when its yield increases by 100 bps is best described as:

A. less than the change in price for a decrease in its yield by 100 bps.
B. the same as the change in price for a decrease in its yield by 100 bps.
C. greater than the change in price for a decrease in its yield by 100 bps.

Hide Solution

Solution:

A is correct. The price–yield relationship for most bonds is convex, such that the increase in the price of a bond is greater for yield decreases than a decrease in the price for the same yield increase.

Discuss Filter

Discussion

What do you want to discuss... (required)

No messages to display

© 2025 CFA Institute. All Rights Reserved.

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search